

K917 — Two peak-convergence catches (both at “ready to call it won”), a re-tier, and Casey’s steer: BACK TO THE WAVES. CATCH 2 (mine, retracted): K916 framed the b-quark 6% as a CONFIRMING “writing degrades outward” signature — WRONG SIGN. Pochhammer predicts $m_b/m_d=840$; observed ~ 900 (ABOVE); the D_3 correction only pushes it BELOW 840 (away from data). The b-quark is NOT a clean test (wrong sign + scale-contaminated) — do NOT log as confirmation. CATCH 1 (Grace’s walk-back of her own “placement won,” ratified): “the ground rung sits at the self-dual center” conflated TWO grounds — the domain vacuum (0,0), solid, and the lightest quark GENERATION (a confined fermion mode), not obviously the same point. Competing reading ($\geq$ as physical): confined=massive=discrete states at the c-function POLES (1:3:5), OFF-center $\rightarrow$ nothing at the self-dual point, anchor collapses. So placement is the OPEN CRUX, not won. The real gate (before the $\lambda \rightarrow x$ number): do the quark rungs sit on the TEMPERED tower (ground $\rightarrow$ center, supports the reframe) or the DISCRETE-SERIES poles at 1:3:5 (off-center, kills it)? CASEY’S STEER: unless remarkable success, BACK TO THE WAVES — don’t over-focus the down-mass; park it as research; resume the broader N1 partition waves.

Keeper | 2026-07-25 Sat | Owned my b-quark wrong-sign; ratified Grace’s placement walk-back; re-tiered the down-quark to placement-OPEN. Per Casey, parking the down-mass as research and going back to the waves.

Catch 2 — own it (my K916 error)

- K916: “b-drift = writing-head signature (exactness degrading outward), predicts the 6%.”
- Grace’s catch: WRONG SIGN. m_b/m_d Pochhammer = 840; observed ~ 900 (ABOVE 840); the D_3 correction pushes BELOW 840 (toward the zero at $\pi/6$) — AWAY from the data. So “degrading outward” does not confirm the b-quark; it mispredicts the direction.
- Retract: the b-quark is NOT a clean test (wrong-signed + m_b scale-contaminated). Do not log as confirmation. m_s/m_d is the clean decider; m_b stays flagged.
- Discipline: a poetically-appealing story that clicks, caught in my own relay by the foreground team. Exactly the peak-convergence trap. Owned.

Catch 1 — ratify Grace’s walk-back (the load-bearing one)

- Grace’s prior “placement won”: the confined ground rung sits at the self-dual center (= vacuum), target-innocent.
- Her walk-back (looked harder): this conflated the DOMAIN VACUUM (0,0) — solid — with the lightest quark GENERATION (a confined fermion mode). Not obviously the same point.
- Competing reading ($\geq$ as physical): confined = massive = discrete-series states at the c-function POLES (imaginary parts 1:3:5), which are OFF the self-dual center. Under this reading nothing sits at the self-dual point $\rightarrow$ the anchor collapses.
- $\square$ placement is the OPEN CRUX, not won. The real rep-theory gate (logically BEFORE the $\lambda \rightarrow x$ flat-window number): do the quark rungs occupy the TEMPERED tower (ground $\rightarrow$ center, supports Casey’s self-dual reframe) or the DISCRETE-SERIES poles at 1:3:5 (off-center, undermines it)? This decides target-innocent

vs target-aware.

- Ratified. Grace corrected her own over-claim at peak convergence — the discipline working, in both directions (she also noted not to over-correct: it's UNDECIDED, not refuted).

Re-tier the down-quark (honest, corrected)

- PLACEMENT: OPEN CRUX (tempered-tower-vs-poles rep-theory question) — NOT won (K916's "target-innocent win" walked back with Grace).
- EXACT-20 ($\lambda \rightarrow x$ scale): downstream of placement; only meaningful IF the tempered-tower reading holds.
- b-quark: NOT a clean test (wrong-signed, scale-contaminated) — flagged, not confirmation.
- What stands: the self-dual center = the vacuum (0,0), solid (Grace); the c-function is explicit (poles at $i/4$ on the critical line); the RH shared-object connection at its tier. The DOWN-QUARK VALUE ($m_s/m_d=20$, Cabibbo) stays CANDIDATE — now with TWO ordered gates: (1) tempered-tower-vs-poles placement; (2) $\lambda \rightarrow x$ scale. Both parked as research (Casey).

Casey's steer — BACK TO THE WAVES (the refocus)

"After this, unless we have remarkable success, let's go back to our waves. We will remember to research the down mass and our progress, but let's not over-focus there right now." - The down-quark consumed waves $3/4$ + the target-innocence/self-dual/writing-head/ $\lambda \rightarrow x$ threads. It's ONE candidate row. The real deliverable is the partition THEOREM (the capstone). - PARK the down-mass as a research lane (the two ordered gates: tempered-tower-vs-poles placement, then $\lambda \rightarrow x$ scale). Remember it; don't over-focus. - RESUME the broader N1 waves — the partition theorem's completion: the buckets (containment #2, moduli #3, runners #4), lives-there (#6, the mixing-values species), the SM table (#7), exhaustiveness (#8). The full task board (team_prompt_2026-07-25n) is the frame.

What's banked / unchanged (the durable state)

- The color partition-LINE THEOREM (F693): banked, in the flagship. The capstone — untouched by all the down-quark back-and-forth.
- Leptons: proven structural (K902).
- Vertical Conjecture C: proved (mass=overlap, the scale).
- Containment (genus): theorem (F595).
- Nothing false banked across the arc — including my b-quark wrong-sign (caught) + Grace's placement over-claim (walked back). The discipline held in both directions.

Directions — the next wave (partition completion; down-mass parked)

- ★ THE WAVE (foreground team, task board -n): advance the buckets toward EXHAUSTIVENESS — #2 containment table (Lyra+Grace), #3 moduli enumeration + prove finite (Grace+Keeper), #4 runners (Grace+Elie); #6 lives-there / mixing-values species (Lyra); #7 SM table (all); #8 exhaustiveness + honesty (Keeper). The row-by-row partition completion.
- ★ PARKED RESEARCH (down-mass, low priority): (1) tempered-tower-vs-poles — do the confined quark rungs sit on the tempered tower or the 1:3:5 discrete-series poles? (Lyra rep-theory + Grace sourcing.) (2) the $\lambda \rightarrow x$ scale (Elie), only if (1) supports the tower reading. Remember, don't over-focus.
- KEEPER: b-quark retraction owned; placement re-tiered to open crux; down-mass parked; the wave refocus on the partition theorem; #8 exhaustiveness standing.

— Keeper K917, 2026-07-25. Two catches: (2, mine, retracted) the b-quark 6% is WRONG-SIGNED, not a confirming signature (Pochhammer 840, obs ~900 above, D_3 pushes below); (1, Grace's walk-back, ratified) "placement won" conflated the domain vacuum with the quark generation — the real crux is tempered-tower ($\rightarrow$ center) vs discrete-series poles at 1:3:5 (off-center), UNDECIDED. Down-quark re-tiered: placement OPEN, then $\lambda \rightarrow x$; value candidate. Casey's steer: BACK TO THE WAVES — park the down-mass as research (the two ordered gates), resume the broader N1 partition waves (buckets, lives-there,

table, exhaustiveness). Capstone (color partition-line) + leptons + vertical Conjecture C unchanged. See
[[Keeper_K916_Grace_WINS_target_innocence_down_is_vacuum]], [[Keeper_K906_N1_DEEP_PUSH_CHARTER]].